COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 | Automating Management Systems	**SECTION:**	CPE31S2
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Prelim Exam Skills

1. PROCEDURE OUTPUT

Hostname	Operating System	Address
workstation	Ubuntu	192.168.56.127
server1	Ubuntu	192.168.56.128
masongsong	CentOS	192.168.56.130

PART A:
Submit a screenshot of the following:

- Ansible version

```
masongsong@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/masongsong/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/masongsong/.ansible/collections:/usr/share
/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
masongsong@workstation:~$
```

- SSH connectivity

```
masongsong@workstation:~$ ssh masongsong@192.168.56.128
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

87 updates can be applied immediately.
69 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:55:34 2026 from 192.168.56.127
masongsong@server1:~$
```

```
masongsong@workstation:~$ ssh masongsong@192.168.56.130
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 08:14:36 2026
[masongsong@localhost ~]$ exit
```

PART B:

Demonstrate the following:

- inventory.yaml

masongsong@localhost:~

```
  GNU nano 7.2                    inventory.yaml
all:
  children:
    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.127
        ubuntu2:
          ansible_host: 192.168.56.128
    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.130
```

- Successful inventory verification

```
masongsong@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.56.130"
            },
            "ubuntu1": {
                "ansible_host": "192.168.56.127"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.128"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu",
            "centos"
        ]
    },
    "centos": {
        "hosts": [
```

- Successful ping results

```
masongsong@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
masongsong@workstation:~/ansible-lab$
```

PART C:

Demonstrate successful execution of the metagroup.

```
masongsong@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.
yaml
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

```
masongsong@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inven
tory.yaml
  hosts (3):
    ubuntu1
    ubuntu2
    centos1
masongsong@workstation:~/ansible-lab$
```

PART D:

Demonstrate successful use of variables.

```
masongsong@centos:~

TASK [Gathering Facts] *********************************************************
***
ok: [ubuntu1]
ok: [ubuntu2]
ok: [centos1]

TASK [Show Operating System] ***************************************************
***
ok: [ubuntu1] => {
    "msg": "Host ubuntu1 is running Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Host ubuntu2 is running Ubuntu"
}
ok: [centos1] => {
    "msg": "Host centos1 is running CentOS"
}

PLAY RECAP *********************************************************************
***
centos1                    : ok=2    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
ubuntu2                    : ok=2    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0

masongsong@workstation:~/ansible-lab$
```

PART E:
Demonstrate:

- Successful playbook execution

```
masongsong@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml play
book.yaml

PLAY [First Ansible Playbook] ***********************************************
***

TASK [Gathering Facts] ******************************************************
***
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Ping all managed nodes] ***********************************************
***
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [Display Hello World] ***************************************************
***
ok: [ubuntu1] => {
    "msg": "Hello World from ubuntu1"
}
ok: [ubuntu2] => {
    "msg": "Hello World from ubuntu2"
}
ok: [centos1] => {
    "msg": "Hello World from centos1"
```

- Successful task completion on all managed nodes

```
TASK [Display Operating System] ********************************************
***
ok: [ubuntu1] => {
    "msg": "Operating System: Ubuntu"
}
ok: [ubuntu2] => {
    "msg": "Operating System: Ubuntu"
}
ok: [centos1] => {
    "msg": "Operating System: CentOS"
}

PLAY RECAP *********************************************************************
***
centos1                    : ok=4    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
ubuntu1                    : ok=4    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
ubuntu2                    : ok=4    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers that contains only:

- ubuntu1
- centos1

masongsong@centos:~

```
 GNU nano 7.2                        servers.yaml
all:
  children:
    ubuntu:
      vars:
       ansible_user: masongsong
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.128
          os_type: Ubuntu
    centos:
      vars:
         ansible_user: masongsong
      hosts:
         centos1:
          ansible_host: 192.168.56.130
          os_type: CentOS
    linuxservers:
      children:
        ubuntu:
        centos:
```

Create a playbook named challenge.yaml that displays the message:

masongsong@centos:~

```
  GNU nano 7.2                          challenge.yaml
---
- name: Display Variables
  hosts: linuxservers

  tasks:

    - name: Show Welcome Message
      debug:
        msg: "Welcome to Ansible Laboratory"
```

Run the playbook and demonstrate that only hosts in the servers group receive the message.

```
masongsong@workstation:~/ansible-lab$ ansible-playbook -i servers.yaml challe
nge.yaml

PLAY [Display Variables] ********************************************************
***

TASK [Gathering Facts] **********************************************************
***
ok: [ubuntu1]
ok: [centos1]

TASK [Show Welcome Message] *****************************************************
***
ok: [ubuntu1] => {
    "msg": "Welcome to Ansible Laboratory"
}
ok: [centos1] => {
    "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP **********************************************************************
***
centos1                    : ok=2    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
ubuntu1                    : ok=2    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
```

Guide Questions

1. What is the role of the Control Node in Ansible?

-It is where you set or create the playbooks, inventory, and other files that ansible can automate.

2. Why is an inventory file required?

-It shows which hosts/ipaddresses are going to be used/pinged to be used in the playbook

3. What is the purpose of a metagroup?

-It displays that everything in the inventory file is connected to the ssh server.

4. How do variables simplify Ansible administration?

-When creating ansible files such as playbooks with the use of variables, you are able to reuse said playbooks in most automated works.

5. What is the difference between an ad hoc command and a playbook?

-Ad hoc commands are just commands that are usually inputted in a single line only, while a playbook allows for multiple commands in one file, just like a bash script.

Conclusion:

I have learned a lot from this PrelimExam, mainly the setup of ansible and how to create inventories and playbooks to be able to use in ansible administration. It was quite hard to notice the indentations whenever an error is present, but after fixing said errors, the setup/tutorial went by pretty fast and I was able to understand key concepts like variables, indentations, commands, and functions. All in all I had fun in trying to figure out the errors of my code, as well as implementing my knowledge to perform the supplementary activity. I hope to further invest my time on this topic to fully understand its key fundamentals.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Used to identify errors in the code.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **ChatGpt was used to identify the codes and explain why it was wrong.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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